DESIGN AND DEVELOPMENT OF A CO-CURRICULAR MANAGEMENT SYSTEM WITH INTELLIGENT REPORTING AND AI-BASED ACTIVITY RECOMMENDATIONS

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THE RESEARCHERS

ABSTRACT

Title: **DESIGN AND DEVELOPMENT OF A CO-CURRICULAR MANAGEMENT SYSTEM WITH INTELLIGENT REPORTING AND AI-BASED ACTIVITY RECOMMENDATIONS**

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Co-curricular activities play an important role in developing students' leadership, communication, teamwork, and organizational skills. These activities help students gain experience, gain confidence, and build connections in school. However, managing data for student organizations, events, attendance, and other records in manual processes can be slow, inefficient and prone to errors. These challenges make it really difficult to

organize student information, manage student participation, and submit reports on time. As the number of student organizations and activities continues to grow, manual recording becomes more difficult to manage and it may affect the overall efficiency of the institution. To address these challenges, this s aims to design and develop a Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations. It is designed to help students, faculty, and administrators to manage co-curricular activities in a organized way. It includes features such as club registration, membership management, event logs, officer elections, officer elections, volunteer monitoring, and report generation. It also provides intelligent reporting to help administrators to create reports more quickly and accurately. In addition, the AI-based activity recommendation feature suggests activities to students based on their interests and previous participations to let them have the encouragement and involvement in the school programs.

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CHAPTER 1

INTRODUCTION

1.1 Project Background and Motivation

Co-curricular activities are an important part of student life because they help students develop skills that go beyond what they learn inside the classroom. By joining clubs, organizations, leadership programs, and school events, students can improve their leadership, communication, teamwork, responsibility, and time management skills. Recent studies continue to support the value of student participation in co-curricular activities. Yeo et. al., (2023) found that participation in school-based co-curricular activities promotes students' confidence, communication skills, teamwork, and sense of belonging when students are motivated to participate. Likewise, Caparros (2023) reported that well-implemented extracurricular and co-curricular programs contribute to school effectiveness by improving student engagement and creating a more active learning environment. Furthermore, Diano et al., (2025) concluded that strong school support for student clubs and organizations significantly increases student participation, satisfaction, and overall involvement in extracurricular activities. These studies show that co-curricular activities play an important role in helping students become active, responsible, and

well-rounded individuals while supporting both their academic and personal development.

At Bestlink College of the Philippines (BCP), co-curricular activities are organized to encourage student participation and holistic development. However, as the number of students, organizations, and school activities continues to increase, managing co-curricular records has become more challenging. Information such as organization memberships, attendance, event participation, volunteer hours, and accomplishment reports is often handled through manual or traditional methods. These processes can be time-consuming, prone to errors, and difficult to manage, especially when preparing reports or monitoring student participation. Because of these challenges, BCP needs a centralized Co-Curricular Management System that can securely store records, simplify administrative tasks, and provide accurate information whenever needed. The proposed system will also integrate Artificial Intelligence through a content-based filtering algorithm, which recommends activities by analyzing a student's interests and previous participation, and a rule-based recommendation algorithm, which applies predefined conditions such as academic program, year level, organization membership, leadership experience, and completed volunteer hours to suggest suitable activities. These algorithms will help students discover relevant opportunities while assisting administrators in promoting

activities to the appropriate participants, making co-curricular management more efficient and data driven.

To address these challenges, this study proposes the Design and Development of a Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations for Bestlink College of the Philippines. The proposed system will provide a centralized platform for managing student organizations, memberships, club officer records, events, attendance, volunteer hours, achievement records, and accomplishment reports. It will also feature an Intelligent Reporting module that automatically generates organized and accurate reports, reducing the time spent on manual report preparation and helping administrators make faster and more informed decisions. By digitizing co-curricular records, the system aims to improve efficiency, data accuracy, and transparency in managing student activities.

In addition, the proposed system will include an AI-Based Activity Recommendation feature that provides personalized activity suggestions based on students' interests, previous participation, and organizational involvement. Instead of manually searching for activities that match their interests, students will receive recommendations that are more relevant to their profiles, encouraging greater participation in school programs. This feature can also help student organizations reach participants who are

more likely to engage in their activities, increasing the success of school events and improving student involvement. Overall, the proposed Co-Curricular Management System aims to modernize the administration of co-curricular activities at Bestlink College of the Philippines by providing a more efficient, organized, and intelligent platform that benefits both students and school administrators.

1.2 Problem Statement

The current process of managing organization memberships, attendance, event participation, volunteer hours, and reports is time-consuming and may lead to errors such as missing records, duplicate entries, and delays in report preparation. These challenges make it difficult for administrators, faculty advisers, and student leaders to monitor student participation and manage co-curricular activities efficiently. In addition, students may not always be aware of activities that match their interests, which can result in lower participation in school program.

General Problem

The general problem of this project is to determine how a Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations can help improve the management of co-

curricular activities and student organizations of Bestlink College of the Philippines.

Specific Problems

1. How can the system improve the efficiency and organization of managing co-curricular activities at Bestlink College of the Philippines?
2. How can the Co-Curricular Management System improve the management of student organizations, memberships, attendance, and event records at Bestlink College of the Philippines?
3. How can the Intelligent Reporting feature help administrators generate accurate and timely reports?
4. How can AI-Based Activity Recommendations encourage students to participate in co-curricular activities that match their interests and previous involvement?

1.3 Project Vision and Scope

Project Vision

This project focuses on the development and implementation of the Co-curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations which can help the Supreme Student Council (SSC) in managing the co-curricular activities more efficiently. It also aims

to support the Supreme Student Council (SSC) by providing Intelligent Reporting for faster and more accurate report generation and AI-Based Activity Recommendations that suggests activities suitable for students based on their interests and previous participation. The system is expected to make managing co-curricular activities faster and more organized, while encouraging more students to take part in school programs of Bestlink College of the Philippines.

1.3.1 In-Scope

The proposed system has the following modules:

- Club Registration Portal for managing the registration, verification, and updating of student organization information.
- Student Club Memberships for managing student membership applications, organization affiliations, and member records.
- Club Officer Elections for managing officer nominations, voting processes, and election results of student organizations.
- Event and Activity Logs for recording, monitoring, and maintaining records of organization events and activities.
- Attendance Tracker for recording and monitoring student participation and attendance in club activities and events.

- Club Achievement Records for storing and managing organization achievements, awards, certificates, and recognitions.
- Budget Requests for managing organization budget submissions, approval processes, and financial request tracking.
- Inter-Club Communication for facilitating announcements, updates, and communication between student organizations and administrators.
- Volunteer Hour Tracking for recording and monitoring student volunteer hours and participation in organizational activities.
- Intelligent Reporting for generating accurate data
- AI-Based Activity Recommendations for recommending activities or events from the students' past participation and interests

1.3.2 Out-of-Scope

The proposed Co-Curricular Management System focuses only on managing student organizations, memberships, events, attendance, volunteer hours, achievements, budget requests, and AI-based activity recommendations. It does not include enrollment, grading, tuition payments, library services, alumni records, inventory management, or other school management functions. The system is intended only for co-curricular activities at Bestlink College of the Philippines.

1.4 Objectives and Goals

This study aims to develop a Co-Curricular Management System for Supreme Student Council of Bestlink College of the Philippines that will help manage student organizations, activities, and records in one system. Specifically, it aims to:

1. Develop a centralized system for managing student organizations, co-curricular activities, and related records in one platform.
2. Provide proper user access by giving each user the features that match their assigned role in the system.
3. Make club management easier through club registration, student membership, officer elections, and club directory features.
4. Improve the management of activities by providing features for event management, attendance tracking, club achievements, and volunteer hours.
5. Generate organized reports about club memberships, activities, attendance, achievements, and volunteer hours through the Intelligent Reporting feature.
6. Recommend suitable activities to students using AI based on their interests and previous participation.

7. Make co-curricular records easier to access and manage by keeping important information in one centralized system.
8. Reduce manual work and improve the management of co-curricular activities by providing a simple, organized, and user-friendly system for students, officers, administrators, and the Supreme Student Council.

1.5 Significance and Relevance

Bestlink College of the Philippines (BCP). The system will provide a centralized platform for managing and monitoring co-curricular activities, improving the efficiency and organization of student organization records and events.

Supreme Student Council (SSC). As the primary client of the system, the SSC will benefit from a centralized platform for managing student organizations, memberships, officer elections, events, attendance, volunteer hours, budget requests, and reports, making administrative tasks faster and more organized.

Student Organizations. The system will help student organizations manage their members, conduct officer elections, organize events, track attendance, submit budget requests, and maintain achievement records in one platform.

Students. The system will make it easier for students to join organizations, participate in activities, monitor their volunteer hours, and receive AI-based activity recommendations based on their interests and previous participation.

Faculty Advisers. The system will allow faculty advisers to monitor organization activities, attendance, and student participation, making it easier to supervise the organizations they handle.

Future Researchers. This study may serve as a reference for future researchers who plan to develop or enhance co-curricular management systems or implement intelligent reporting and AI-based recommendation features in educational institutions.

1.6 Definition of Terms

Agile Scrum Framework. A popular, lightweight Agile framework used by teams to structure, manage, and deliver complex work through short, iterative cycles called sprints

AI-Based Activity Recommendations. A feature that recommends co-curricular activities to students based on their interests and previous participation.

Centralized Platforms. A single, unified system.

Co-Curricular Management System. A web-based system used to manage organizations, memberships, events, attendance, volunteer hours, achievements, and reports.

Enterprise Architecture. A master plan or blueprint that helps a company match its business goals with its technology.

Intelligent Reporting. The system's capability to automatically generate reports, summaries, and dashboards based on co-curricular activity data.

Microservices Architecture. A design method where a large application is built as a set of small, independent services

Monolithic Architecture. A traditional software design where all parts of an application—like the user interface, business logic, and database access—are combined into a single, unified codebase and deployed as one single unit.

Internet of Things (IoT). A large network of physical objects—like devices, vehicles, and home appliances—that have sensors and software. These objects connect to the internet to collect and share data without needing help from people

Student Organization. Recognized clubs and organizations whose information, memberships, and activities are managed within the system.

1.7 Structure of the Document

Chapter 1 – Introduction presents the background and motivation of the study, problem statement, project vision and scope, objectives, significance and relevance, definition of terms, and the overall structure of the document.

Chapter 2 – Related Literature, Studies, and Technical Background reviews related literature, studies, and existing systems relevant to the proposed Co-Curricular Management System. It discusses the Agile Scrum methodology, emerging technologies and standards such as Artificial Intelligence (AI), Internet of Things (IoT), Data Analytics and Business Intelligence, Polyglot Persistence, Cybersecurity, Data Privacy, Software Quality Standards, Cloud Computing, and Edge Computing. It also presents DevOps Culture and CI/CD practices, Enterprise Architecture and System Integration, the Conceptual Framework, and the Theoretical Paradigm that support the development of the proposed system.

Chapter 3 – Methodology and Project Management describes the research methodology and project management approach used in the study. It discusses the adapted Agile Scrum Framework, including team roles and responsibilities, Scrum events, Scrum artifacts, and development toolstack. It also presents the architectural design decisions, development

and quality assurance methodology, DevOps practices, testing strategy, and the innovation framework applied throughout the project.

Chapter 4 – System Design, Development, and Implementation presents the requirements engineering process, stakeholder analysis, system requirements, user stories, and use cases. It also discusses system architecture, business process models, application architecture, database design, technology architecture, infrastructure, development journey, implementation strategy, testing and quality assurance reports, user acceptance testing, system evaluation, validation of objectives, and project retrospective.

Chapter 5 – Conclusion summarizes the entire project, presents the conclusions based on the findings, highlights the project's contributions, and provides recommendations for future enhancements, system implementation, and future researchers.

CHAPTER 2

RELATED LITERATURE, STUDIES, AND TECHNICAL BACKGROUND

2.1 Review of Related Literature

This chapter presents the related literature and studies conducted by previous researchers that are relevant to the present study. These sources provide supporting information, concepts, and findings that help establish the foundation of the research. Furthermore, this chapter aims to deepen the understanding of the study by identifying existing knowledge, research gaps, and ideas that are related to the development of the proposed system.

FOREIGN

According to Wang (2025) that traditional student management systems mainly focus on storing student records and often lack intelligent features that can analyze student data. The study introduced an AI-powered student management system that uses machine learning to monitor student performance and identify learners who may need academic support. The results showed that AI can improve decision-making by providing accurate predictions and timely interventions for students. This study is relevant to the proposed research because both

systems integrate artificial intelligence to improve student management processes. While Wang's research focuses on predicting academic performance, the proposed Co-Curricular Management System applies AI to recommend co-curricular activities based on students' interests and participation history. It also includes an Intelligent Reporting feature that helps administrators generate reports more efficiently, making the system useful not only for managing records but also for encouraging student participation in school activities.

Li (2025) stated that educational institutions need intelligent systems that can organize student information and provide personalized support. The study developed an AI-driven student management platform that combines student records, competency profiles, and learning resources into one system. The findings revealed that the platform improved data management, student monitoring, and personalized services for learners. This study supports the proposed research because both aim to improve the way student information is managed using intelligent technologies. Similar to Li's system, the proposed study centralizes student records into one platform. However, instead of focusing on learning resources and competencies, the proposed system concentrates on managing co-curricular activities while using AI to recommend suitable school programs based on students' interests and previous participation.

According to Zhang et al. (2025) an intelligent student management system that combines academic records, behavioral data, and other student information using deep learning techniques. The researchers found that the system improved management efficiency and increased the accuracy of personalized recommendations compared to traditional systems. They concluded that intelligent technologies could provide better support for educational management and student development. This study is closely related to the proposed research because both use intelligent technologies to improve educational management and provide personalized recommendations. Although Zhang et al. focused on overall student management through deep learning, the proposed study applies intelligent recommendations specifically to co-curricular activities. It also includes Intelligent Reporting to help school administrators monitor participation and generate reports more quickly and accurately.

Alfredo et al. (2023) reviewed more than one hundred studies on learning analytics and artificial intelligence in education. They found that AI systems are becoming more effective in supporting student learning, monitoring progress, and assisting educators in decision-making. However, the review also emphasized the importance of involving teachers and students during system development to ensure that AI remains reliable, ethical, and user-friendly. This literature is valuable to the

proposed study because it highlights the importance of designing AI systems that are practical and easy to use. In the same way, the proposed Co-Curricular Management System is designed for both students and administrators by providing an organized interface, intelligent reporting, and AI-based activity recommendations that support better participation and decision-making while remaining user-friendly.

Katsaros et al. (2026) discussed how extracurricular activities contribute to students' academic performance, leadership, teamwork, and personal development. The review also explained that artificial intelligence and machine learning can help educational institutions analyze participation data and improve decision-making regarding student engagement. The authors suggested that AI has great potential in supporting educational management, especially when combined with extracurricular information. This study strongly supports the proposed research because it emphasizes the importance of using AI to improve the management of co-curricular and extracurricular activities. Building on this idea, the proposed Co-Curricular Management System uses machine learning to analyze students' interests and participation history to recommend relevant activities. It also provides Intelligent Reporting that enables administrators to monitor student involvement, evaluate

participation trends, and make better decisions when planning future co-curricular programs.

LOCAL

According to Autor and Esguerra (2025), managing students' experiential learning and co-curricular activities through manual processes often results in delayed transactions, incomplete records, and difficulty in monitoring student participation. To address these issues, they developed Transcript++, a web-based system that digitizes the recording, verification, and evaluation of students' experiential and co-curricular activities. The researchers found that the system improved documentation, simplified record verification, and enabled administrators to monitor students' accomplishments more efficiently. This study is closely related to the proposed research because both aim to replace manual record-keeping with a centralized digital platform for managing co-curricular information. However, the proposed Co-Curricular Management System extends these functions by incorporating Intelligent Reporting and AI-based Activity Recommendations, allowing administrators to generate reports automatically while helping students discover activities that match their interests and participation history.

According to Marzan et al. (2025), manual processes in managing student organizations often lead to misplaced records, delayed report

generation, and ineffective communication among organizations. To solve these challenges, they developed a Club and Academic Organization Management System for Saint Paul University Philippines using the Agile Scrum methodology and evaluated it based on the ISO/IEC 25010 software quality standards. Their findings showed that the system improved organization management, communication, and record-keeping efficiency. This study supports the proposed research because both focus on improving the management of student organizations through digital technology. However, the proposed system goes beyond automation by integrating artificial intelligence to recommend co-curricular activities based on students' interests, organization memberships, and previous participation, making the system more interactive and student-centered.

Educational institutions need digital information systems to improve the management of academic processes and institutional records. They developed a curriculum management system that enables faculty members and school administrators to review, update, and manage curriculum information through a centralized platform, Dela Rosa and Galang (2023). Their findings revealed that the system improved collaboration among stakeholders, minimized manual work, and increased the efficiency of curriculum management. Although their research focuses on curriculum administration rather than co-curricular management, it demonstrates how

centralized information systems can improve school operations and reduce administrative workload. This idea supports the proposed study because it also aims to provide a centralized platform that simplifies the management of co-curricular records while adding Intelligent Reporting and AI-based Activity Recommendations to improve decision-making and student engagement.

According to Argayoso et al. (2015), their developed online management system allows student organizations to submit proposal forms, accomplishment reports, liquidation reports, and evaluation documents electronically. The researchers concluded that digitizing these organizational transactions reduced paperwork, improved processing efficiency, and increased user satisfaction among both students and administrators. This study is relevant to the proposed research because both systems aim to modernize the management of student organizations through digital technology. However, unlike their system, which mainly focuses on online document submission, the proposed Co-Curricular Management System introduces Intelligent Reporting for faster report generation and AI-based Activity Recommendations that help students identify co-curricular activities suited to their interests and previous involvement, providing additional support for both students and administrators.

Eulin et al. (2019) developed a Student Information System that includes modules for student organizations and other student services. The researchers highlighted that centralized information systems improve the storage, retrieval, and management of student records while reducing paperwork, processing time, and data redundancy. They also found that automating student services increased the accuracy, reliability, and accessibility of student information. This study is related to the proposed research because both promote the use of centralized information systems to improve the management of student records and organizational activities. While the Student Information System mainly focuses on storing and managing student information, the proposed Co-Curricular Management System expands these capabilities by incorporating Intelligent Reporting and AI-Based Activity Recommendations, allowing administrators to make informed decisions while encouraging students to participate more actively in co-curricular programs.

2.2 Analysis of Related Studies

The reviewed foreign studies show that educational institutions are continuously adopting artificial intelligence and advanced technologies to improve student management systems. According to Wang (2025), integrating machine learning into student management systems allows schools to analyze student information more effectively and provide timely support for students. Similarly, Li (2025) developed an AI-driven student management platform that organizes student records and delivers personalized services based on students' profiles and competencies. Zhang et al. (2025) also demonstrated that intelligent student management systems can improve management efficiency and generate more accurate recommendations by analyzing different types of student data. These studies are closely related to the proposed research because they show how artificial intelligence can improve decision-making and automate educational processes. However, most of these studies focused on academic management, student performance, and general student services rather than managing co-curricular activities. This creates a gap that the present study aims to address by applying AI specifically to co-curricular management through Intelligent Reporting and AI-based Activity Recommendations.

In addition to intelligent student management systems, other foreign studies emphasized the importance of artificial intelligence in supporting educational management and student development. Alfredo et al. (2023) explained that artificial intelligence and learning analytics help educators monitor student progress, identify learning needs, and make better decisions. Likewise, Katsaros et al. (2026) highlighted that extracurricular and co-curricular activities contribute to students' leadership, teamwork, communication skills, and personal development. The authors also suggested that AI and machine learning can analyze participation records to improve student engagement and recommend suitable activities. These findings support the proposed study because they demonstrate the potential of AI to improve both administrative tasks and student participation. However, these studies mainly discussed the benefits of AI and learning analytics without developing a complete system that combines co-curricular management, intelligent reporting, and personalized activity recommendations. The proposed study intends to fill this gap by integrating these features into a single platform designed specifically for managing co-curricular programs.

The reviewed local studies mainly focused on improving student organization management through digital systems. According to Autor and Esguerra (2025), their Transcript++ system improved the recording,

verification, and monitoring of students' experiential learning and co-curricular activities by replacing manual processes with a web-based platform. Marzan et al. (2025) also developed a Club and Academic Organization Management System that automated organization records, reports, and communication, making administrative tasks more efficient. Similarly, Argayoso et al. (2015) introduced an online transaction processing system that enabled recognized student organizations to submit proposals, accomplishment reports, and other documents electronically. These studies are relevant because they demonstrate how digital technologies can improve the management of student organizations and reduce paperwork. However, their systems mainly focused on digitizing transactions and managing organizational records. They did not include intelligent reporting features or artificial intelligence that could analyze student participation and provide personalized activity recommendations. This limitation provides an opportunity for the proposed study to develop a more intelligent and interactive co-curricular management system.

Other local researchers also emphasized the importance of centralized information systems in educational institutions. Dela Rosa and Galang (2023) found that a curriculum management system improved collaboration among faculty members while reducing the time and effort

required for curriculum management. Likewise, Eulin et al. (2019) explained that a Student Information System improved the storage, retrieval, and accessibility of student records while reducing manual processing. These studies support the proposed research because they prove that centralized digital systems can improve the efficiency and quality of school operations. However, they primarily focused on academic administration and student record management instead of co-curricular activities. In addition, they did not utilize artificial intelligence to generate intelligent reports or recommend activities based on students' interests and participation history. The proposed study addresses these limitations by combining centralized record management with AI-powered features that support both administrators and students.

After comparing the foreign and local studies, it can be observed that both have the common goal of improving educational management through technology. Foreign studies generally focus on integrating artificial intelligence, machine learning, learning analytics, and personalized recommendation techniques into student management systems, while local studies mainly emphasize digitizing school processes, student organization management, and record keeping. Although these studies have contributed significantly to educational technology, there is still a noticeable gap. Most existing systems either focus on academic

management or simply automate manual processes without providing intelligent features that support co-curricular management.

Based on the reviewed literature and studies, the researchers found that only a few systems combine co-curricular management, Intelligent Reporting, and AI-based Activity Recommendations within a single platform. Existing systems commonly manage records, monitor student activities, or automate organizational transactions, but they rarely provide personalized recommendations that encourage students to participate in activities that match their interests and previous involvement. Furthermore, many systems still require administrators to manually analyze participation records when generating reports and planning future programs. To address these gaps, the proposed study, "Design and Development of a Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations," aims to develop a centralized system that not only manages co-curricular activities but also uses artificial intelligence to generate intelligent reports and recommend suitable activities for students. Through these features, the proposed system is expected to help administrators make better decisions, reduce administrative workload, and encourage greater student participation in co-curricular programs.

2.3. Agile Scrum Methodology

The proposed Design and Development of a Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations will be developed using the Agile Scrum Methodology. Agile Scrum is an iterative and incremental software development approach that emphasizes flexibility, collaboration, and continuous improvement, making it suitable for developing a system with intelligent reporting and AI-based recommendation features.

The methodology focuses on developing functional system modules through short development cycles called sprints while continuously gathering feedback from the Supreme Student Council, the client of the system. This approach enables the development team to identify necessary improvements, respond to changing requirements, and refine system features throughout the development process. By adopting Agile Scrum, the proposed system ensures that its functionalities remain aligned with the co-curricular management needs of Bestlink College of the Philippines.

2.4 Emerging Technologies, Intelligent System and Intelligent Standards in Modern Information Systems

Modern information systems use different technologies to make tasks easier, faster, and more accurate. Some of these technologies include artificial intelligence (AI), cloud computing, data analytics, APIs, and microservices architecture. These technologies help organizations manage information efficiently, automate repetitive tasks, and provide users with reliable services.

An intelligent system uses AI to assist users in making better decisions based on available data. In this study, the proposed Co-Curricular Management System includes two intelligent features: Intelligent Reporting, which automatically generates reports from the collected data, and AI-Based Activity Recommendations, which suggest co-curricular activities that match a student's interests and previous participation. These features help reduce manual work and encourage students to become more involved in school activities.

The proposed system also follows modern development practices by using microservices architecture and API-based communication. These approaches make the system easier to maintain, improve, and expand in the future while providing a reliable and organized platform for managing co-curricular activities at Bestlink College of the Philippines.

2.4.1 Microservices Architecture

The Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations is designed using a microservices architecture, where the system is divided into independent services that communicate through APIs. Each major module, Club Registration Portal, Student Club Memberships, Club Officer Elections, Event and Activity Logs, Attendance Tracker, Club Achievement Records, Budget Requests, Inter-Club Communication, Club Directory, Intelligent Reporting, and AI-Based Activity Recommendations, operates as a separate service that can be developed, tested, deployed, and maintained independently.

This architecture improves the system's flexibility, scalability, and maintainability. Updates or enhancements to one service, such as the AI-Based Activity Recommendations or Intelligent Reporting feature, can be implemented without affecting the other modules. It also allows additional features and services to be integrated in the future without requiring major changes to the overall system architecture.

The microservices architecture supports the Agile Scrum development methodology by allowing different services to be developed simultaneously during sprint cycles. It also improves system reliability, as issues encountered in one service do not interrupt the operation of the

other services. Overall, this architecture enables the proposed system to efficiently support the co-curricular management needs of the Supreme Student Council and Bestlink College of the Philippines, while remaining adaptable for future enhancements and system integration.

2.4.2 Artificial Intelligence (AI)

In this study, AI is used through the AI-Based Activity Recommendations feature of the proposed Co-Curricular Management System. The system analyzes a student's interests, previous participation, and available activities to recommend organizations and events that may be relevant to them. This feature helps students discover suitable co-curricular activities and encourages greater participation in school organizations. By using AI, the system provides a more personalized and efficient experience for students while assisting the Supreme Student Council in promoting student engagement.

2.4.3 Internet of Things (IoT)

The proposed Co-Curricular Management System does not use Internet of Things (IoT) technology because its main purpose is to manage and organize co-curricular information through a web-based platform. The system focuses on recording student data, generating intelligent reports, and providing AI-based activity recommendations using stored information instead of collecting data from physical devices or sensors. Since the

system does not require real-time data from connected hardware, IoT is not necessary for achieving the objectives of the study.

2.4.4 Data Analytics and Business Intelligence

Data Analytics and Business Intelligence (BI) are technologies that help organizations collect, organize, and analyze data to support better decision-making. While Data Analytics focuses on identifying patterns and trends from data, Business Intelligence presents the information through reports, dashboards, and visual summaries that are easier to understand.

Data Analytics and Business Intelligence are applied through the Intelligent Reporting feature. The system collects data from student organizations, memberships, attendance, volunteer hours, and events, then organizes the information into reports that can help the Supreme Student Council monitor participation, evaluate activities, and make informed decisions. These technologies help reduce manual work, improve report accuracy, and provide useful information for planning future co-curricular programs.

2.4.5 Polyglot Persistence

Polyglot Persistence is the practice of using different data storage technologies within a single system, where each database is selected based on the type of data it manages. Instead of storing all information in one database, this approach uses the most suitable storage solution for different system requirements, improving performance, flexibility, and scalability.

The system primarily uses a relational database to manage structured data such as student records, organization information, memberships, attendance, volunteer hours, and event details. As the system is built using a microservices architecture, it can support the integration of different data storage technologies in the future if needed, allowing each service to use the database that best fits its purpose. This approach makes the system more adaptable and easier to expand as new features are introduced.

2.4.6 Cybersecurity

Cybersecurity refers to the protection of information systems, networks, and data from unauthorized access, cyberattacks, and other security threats. In the Philippine context, securing digital platforms against malicious digital attacks is heavily guided by the Cybercrime Prevention Act of 2012 (Republic Act No. 10175). This law mandates that educational

institutions strictly protect their network infrastructure and sensitive databases from unauthorized breaches, data interference, and illegal access.

To comply with these national security standards, cybersecurity in the proposed Co-Curricular Management System is implemented through user authentication, password hashing, HTTPS/SSL encryption, input validation, and SQL injection prevention. User authentication ensures that only registered users can access the system, while password hashing protects user credentials from unauthorized disclosure. HTTPS/SSL encryption secures data transmitted between users and the system.

Furthermore, input validation combined with SQL injection and Cross-Site Request Forgery (CSRF) prevention helps protect the database from malicious attacks. These technical security measures help safeguard the student records, organization information, attendance data, and other sensitive information managed by the Supreme Student Council, ensuring the system remains legally compliant and reliable against cyber threats.

2.4.7 Data Privacy

Data privacy refers to the ethical and legal handling, collection, and management of personal information. In the Philippines, this is strictly governed by the Data Privacy Act of 2012 (Republic Act No. 10173) and the guidelines set by the National Privacy Commission (NPC). The law mandates that all personal data must be processed according to three core principles: transparency, legitimate purpose, and proportionality.

In the proposed Co-Curricular Management System, data privacy is applied by ensuring that student information, organization records, and attendance data are collected only after obtaining the users' clear consent (Transparency). The system guarantees that the collected information is utilized strictly for its intended objective, managing co-curricular activities and generating accurate intelligent reports and will not be shared or used for outside reasons (Legitimate Purpose).

Furthermore, the system enforces the law's requirement for Proportionality through role-based data visibility. This means the system only exposes the exact amount of data necessary for a user's specific role. A regular student can only view their own membership and attendance records. Meanwhile, faculty advisers and the Supreme Student Council (SSC) can only access the specific, limited information required to perform their administrative duties. By strictly adhering to these Philippine privacy

laws, the system maintains user trust and protects the sensitive information of the Bestlink College of the Philippines student body.

2.4.8 Software Quality Standards

Software Quality Standards are established guidelines used to ensure that a system is reliable, maintainable, secure, and performs according to user requirements. In this study, the evaluation of the software is anchored on the ISO/IEC 25010 Software Product Quality Model, which is the international standard for evaluating software systems.

To ensure the Co-Curricular Management System is efficient and reliable, the development follows the core characteristics of this standard. Functional Suitability is achieved by ensuring the system accurately processes club registrations, attendance tracking, and AI-based activity recommendations according to the needs of the Supreme Student Council. Usability is prioritized by designing an intuitive, user-friendly interface that allows students and administrators to navigate the platform with ease.

Furthermore, the system ensures Performance Efficiency by generating Intelligent Reports quickly without lagging, even when handling large amounts of student data. Security is maintained through encryption and strict access controls, while the modular architecture ensures high Maintainability, allowing future developers to easily update the system. By

adhering to the ISO/IEC 25010 standards, the system guarantees a high-quality, professional-grade platform for managing co-curricular activities

2.4.9 Cloud Computing

Cloud computing is a technology that allows users to access applications, data, and computing resources through the internet instead of depending only on a local computer or server. It enables users to store, manage, and retrieve information online, making systems more accessible, flexible, and efficient. Many organizations use cloud computing because it supports remote access, data backup, and reliable storage while reducing the need for expensive hardware.

Cloud computing supports the online management of student organizations, memberships, attendance records, volunteer hours, event information, and reports. Since the system is web-based, authorized users such as the Supreme Student Council, faculty advisers, and students can access the system using an internet connection. This allows users to manage records, monitor activities, and generate reports from different locations and devices.

2.4.10 Edge Computing

Edge computing is a technology that processes data closer to where it is collected instead of sending all the data to a central server or cloud. This helps reduce delays, improve system performance, and provide faster

responses. Edge computing is commonly used in applications that require real-time data processing, such as smart devices, security systems, and Internet of Things (IoT) applications.

Although edge computing is an important technology in modern information systems, it is not directly implemented in the proposed Co-Curricular Management System. The system is designed as a web-based application that processes and stores data through its central server. However, edge computing may be considered as a future enhancement if the system integrates smart devices, such as QR code scanners, biometric devices, or other hardware that require faster local data processing.

2.5 DevOps Culture and CI/CD Practices

DevOps is a software development approach that promotes collaboration between developers and operations teams to deliver software more efficiently. It encourages continuous communication, regular testing, and frequent system updates to improve software quality and respond quickly to user feedback. Continuous Integration (CI) allows developers to combine code changes regularly, while Continuous Delivery/Continuous Deployment (CD) helps release updates and new features in a faster and more reliable manner.

DevOps and CI/CD practices support the Agile Scrum methodology used during system development. As new modules and features are

completed, they can be integrated, tested, and improved throughout each sprint. This approach helps the development team identify and resolve issues early while ensuring that the system remains reliable and meets the requirements of the Supreme Student Council. By applying DevOps and CI/CD practices, the system becomes easier to maintain, enhance, and update as future improvements are introduced.

2.6 Enterprise Architecture & System Integration

Enterprise Architecture (EA) is a framework that helps organizations organize their business processes, information, applications, and technology to achieve their goals. It provides a clear structure for designing and managing information systems, ensuring that different system components work together efficiently. System integration, on the other hand, connects different modules or systems so they can share data and communicate with one another.

Enterprise Architecture helps organize the system modules, including the Club Registration Portal, Student Club Memberships, Club Officer Elections, Attendance Tracker, Event and Activity Logs, Intelligent Reporting, and AI-Based Activity Recommendations. Through system integration, these modules can exchange information, allowing data entered in one module to be reflected in other related modules. This creates a centralized and organized system that helps the Supreme

Student Council manage student organizations and co-curricular activities more efficiently while supporting future system enhancements and integration.

2.7 Conceptual Framework

The conceptual framework of this study is based on the Input-Process-Output (IPO) model, which illustrates the step-by-step flow of the project's development and the system's daily operations.

Input: The input phase consists of two parts. The first includes the hardware, software, and system requirements gathered from the Supreme Student Council (SSC) needed to develop the platform. The second part involves the actual raw data that users will input into the system, such as student profiles, declared interests, club information, budget requests, and scanned QR code attendance records.

Process: The process phase represents both the development methodology and the system's internal functions. For development, this involves the Agile Scrum methodology, which includes requirements analysis, database design, coding, and software testing. Operationally, the process phase includes the system's internal logic specifically, how the Artificial Intelligence (AI) analyzes a student's interests to find matching activities, and how the Data Analytics engine automatically computes attendance and volunteer hours.

Output: The final output is the fully functional, web-based Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations. This includes the generation of accurate visual reports for the SSC and personalized event suggestions for the students, ultimately leading to a more efficient and engaged campus environment at Bestlink College of the Philippines.

2.8 Theoretical Paradigm

This study is anchored on the Technology Acceptance Model (TAM) developed by Fred Davis, which is a widely used theory in information systems research. TAM explains that the success and actual usage of a new technology depend on two primary factors: Perceived Usefulness and Perceived Ease of Use.

Perceived Usefulness refers to the degree to which a user believes that using a system will improve their job performance. In this study, the Intelligent Reporting feature provides high perceived usefulness for the Supreme Student Council (SSC) and faculty advisers by eliminating manual paperwork, automating attendance tracking, and generating instant reports. For the students, the AI-Based Activity Recommendations are highly useful because they automatically filter out irrelevant events and suggest clubs that perfectly match their personal interests.

Perceived Ease of Use refers to how effortless the system is to operate. To achieve this, the proposed system is designed as a web-based platform with an intuitive, user-friendly interface. By using simple QR codes for attendance and organizing the modules logically, the system ensures that users of all technical skill levels can navigate it without confusion. According to this theoretical paradigm, by ensuring the system is both highly useful for administrative tasks and incredibly easy for students to use, the Bestlink College of the Philippines community will readily accept, adopt, and continuously use the platform.

CHAPTER 3

METHODOLOGY AND PROJECT MANAGEMENT

3.1 The Adapted Agile Scrum Framework

The researchers used the Adapted Agile Scrum Framework to develop the Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations. The system was designed to help manage the school's co-curricular activities, student organizations, attendance records, and event information while generating reports more efficiently. The Agile Scrum Framework was selected because it provides a flexible and iterative approach to software development, allowing the researchers to complete the project through short development cycles called sprints. This framework encourages continuous planning, development, testing, and improvement based on stakeholder feedback and changing project requirements. According to Schwaber and Sutherland (2020), Scrum enables development teams to deliver valuable software incrementally while promoting transparency, inspection, and adaptation throughout the development process.

Using the Agile Scrum Framework allowed the researchers to develop the system step by step instead of building the entire project at once. Each sprint focused on a specific set of system features,

including the Club Registration Portal, Student Club Memberships, Attendance Tracker, Event Management, Intelligent Reporting, and AI-Based Activity Recommendations. At the beginning of every sprint, the researchers conducted sprint planning to determine the tasks, priorities, and expected deliverables. The planned features were then designed, developed, and tested before proceeding to the next sprint. This iterative development process made it easier to identify issues early, apply necessary improvements, and ensure that each completed feature met the project's objectives. Pressman and Maxim (2020) explained that Agile development promotes continuous customer collaboration, rapid delivery of functional software, and the ability to adapt to changing requirements throughout the software development life cycle.

The Agile Scrum Framework was adapted to fit the timeline, academic schedule, and development requirements of the researchers. Regular sprint meetings were conducted to monitor the project's progress, discuss challenges encountered during development, and evaluate completed tasks. Feedback from the project owner was incorporated after every sprint to improve the functionality and usability of the system. According to Sommerville (2016), Agile methods encourage close collaboration between developers and stakeholders, allowing software

to evolve continuously through frequent feedback and incremental improvements.

Testing was performed after every sprint to verify that each module functioned correctly before integration with the complete system. Errors and defects identified during testing were corrected immediately to minimize future development issues. Continuous testing also helped ensure that newly developed features did not affect the existing functionalities of the system. This iterative testing process improved the reliability, quality, and maintainability of the software while reducing development risks.

By adapting the Agile Scrum Framework, the researchers were able to organize development activities effectively, respond to requirement changes, and continuously improve the system throughout the project. The framework supported collaboration among the researchers, encouraged continuous evaluation of system features, and ensured that every sprint produced functional outputs. As a result, the researchers successfully developed a Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations that met the identified project objectives while following an organized and systematic software development process. The iterative nature of Scrum also aligns well with software engineering practices that emphasize continuous delivery,

quality assurance, and stakeholder involvement throughout the system development life cycle.

3.1.1 Team Roles and Responsibilities

Table 1: Team Roles and Responsibilities

Role	Assignment	Responsibility
Product Owner / Stakeholder	Mr. Teodolo Jr. N. Panao (SSC President)	Represents the client, defines project requirements, provides feedback, prioritizes features, and approves project deliverables.
Scrum Master	Aris Guiller S. Pangilinan	Supports the development team, facilitates communication, and helps ensure the successful completion of the project.

Project Manager	Lorenz Phillip Q. Llera	Plans, organizes, and monitors project activities to ensure timely completion of project objectives.
Front-End Developer	Brian Joshua P. Tanael	Develops server-side functionalities, manages the database, and integrates system components.
Back-End Developer	Bon Joseph A. Arendain	Develops server-side functionalities, manages the database, and integrates system components.
Systems Analyst	Cheryl C. Gales	Gathers and analyzes requirements, prepares system

		specifications, and validates system processes.

3.1.2 Scrum Events

The Agile Scrum Framework organizes software development into a series of structured events that help teams plan, develop, review, and improve a system throughout the project. These events encourage teamwork, regular communication, and continuous improvement, making it easier to manage tasks and respond to changes during development (Schwaber & Sutherland, 2020). In developing the Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations, the Scrum events served as a guide in completing each system module through a series of sprints. This approach helped ensure that every feature was developed, tested, and improved based on the project requirements and feedback from the client, resulting in a system that met the project's objectives (Pressman & Maxim, 2020).

Table 2: Scrum Events

Scrum Event	Purpose	Application to the Proposed System
Sprint Planning	Defines the goals, tasks, and activities to be completed during a sprint. It helps the team organize the development process based on priorities and requirements.	The development team identifies the modules and features to be developed, such as club registration, attendance tracking, intelligent reporting, and AI-based activity recommendations. The tasks are planned based on the system requirements and project timeline.
Daily Scrum	Provides a short meeting where team members discuss progress, challenges, and plans to maintain	The team regularly discusses the progress of system development, updates completed tasks, addresses

	effective communication and collaboration.	technical issues, and coordinates improvements for the Co-Curricular Management System.
Sprint Review	Evaluates the completed work at the end of each sprint and gathers feedback to determine possible improvements.	The developed system features are presented and reviewed, such as organization management, event monitoring, reports generation, and recommendation features. Feedback from stakeholders is used to improve the system.
Sprint Retrospective	Allows the team to reflect on the completed sprint, identify areas for	The team reviews the development process, identifies problems encountered during

	improvement, and plan better practices for future sprints.	implementation and testing, and applies improvements to enhance the quality and performance of the proposed system.

3.1.3 Scrum Artifacts

The Scrum Artifacts applied in the development of the Design and Development of a Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations served as essential tools for organizing project requirements, planning sprint activities, monitoring progress, and managing the overall development process. According to the Scrum Guide (2020), Scrum artifacts represent work or value and are designed to provide transparency of important information, allowing the Scrum Team and stakeholders to have a shared understanding of the project progress. The three main Scrum artifacts are the Product Backlog, Sprint Backlog, and Increment, which help guide the team in achieving the desired product goals.

The use of Scrum Artifacts helped the development team properly manage system requirements, prioritize features, and track completed

tasks throughout each sprint. The Product Backlog was used to document and organize all required system features, while the Sprint Backlog guided the team in selecting and managing tasks for each development cycle. The completed outputs from every sprint served as Increments, representing the continuous improvement and progress of the proposed system.

Product Backlog

Table 3: Product Backlog

User Story No.	Module/Feature	User Story	Priority	Status
F1	User Authentication and Security	As a user, I want to securely log in and log out so that my account and system information are protected from unauthorized access.	High	Not Yet Done
F2	User Account Management	As an Admin, I want to manage user accounts and roles so that only	High	Not Yet Done

		authorized users can access specific system features.		
F3	Club Registration Portal	As a Student Organization Officer, I want to register our organization in the system so that our club information can be officially recorded.	High	Not Yet Done
F4	Student Club Membership	As a Student, I want to join registered organizations so that I can participate in co-curricular activities.	Medium	Done
F5	Student Club Membership	As an Officer, I want to manage club members so that I can monitor active and inactive members.		Not Yet Done

F6	Club Officer Elections	As an SSC President, I want to manage club officer elections so that the selection of organization leaders is properly conducted and recorded.	High	Not Yet Done
F7	Club Officer Elections	As a Student, I want to participate in officer elections so that I can vote for qualified candidates and contribute to selecting organization leaders.	Medium	Done
F8	Event and Activity Logs	As the SSC President, I want to review and monitor event and activity records so that organization activities are properly tracked and evaluated.	High	Not Yet Done

F9	Attendance Tracker	As an Officer, I want to record student attendance so that participation in activities can be monitored.	High	Not Yet Done
F10	Attendance Tracker	As a Student, I want my attendance records to be tracked so that my participation history is updated.	Medium	Not Yet Done
F11	Club Achievement Records	As an Officer, I want to record organization achievements so that club accomplishments are documented.	Medium	Not Yet Done
F12	Club Achievement Records	As the SSC President, I want to view organization achievements so that I	Medium	Not Yet Done

		can evaluate club performance.		
F13	Budget Requests	As an Officer, I want to submit budget requests for activities so that financial needs can be processed.	High	Not Yet Done
F14	Budget Requests	As the SSC President, I want to review budget requests so that organization expenses can be properly monitored.	High	Not Yet Done
F15	Inter-Club Communication	As an Officer, I want to post announcements so that members receive important organization updates	High	Not Yet Done
F16	Inter-Club Communication	As a Student, I want to receive announcements so that	Medium	Done

		I stay informed about upcoming activities.		
F17	Volunteer Hour Tracking	As a Student, I want my volunteer hours to be recorded so that my participation and contributions are recognized.	High	Not Yet Done
F18	Volunteer Hour Tracking	As an Officer, I want to manage volunteer records so that student involvement can be monitored.	High	Not Yet Done
F19	Club Directory	As a Student, I want to browse registered organizations so that I can discover organizations	Medium	Done
F20	Club Directory	As an SSC President, I want to view all registered	High	Not Yet Done

		organizations so that I can monitor student clubs.		
F21	Intelligent Reporting	As an SSC President, I want to generate automated reports so that I can analyze organization activities and performance.	High	Not Yet Done
F22	Intelligent Reporting	As an Admin, I want to view summarized reports so that I can monitor overall system activities.	High	Not Yet Done
F23	AI-Based Activity Recommendations	As a Student, I want to receive recommended activities based on my interests and previous participation so that I	High	Not Yet Done

		can find suitable events.		

Sprint Backlog

Table 4: Sprint Backlogs

Sprint	Duration	Modules/Features	Deliverables	Status
Sprint 1	June 20 – June 27, 2026	Project Planning and Requirements Analysis	Gathered system requirements, identified users (Admin, SSC President, Officer, Student), defined system scope, and prepared initial documentation.	Completed
Sprint 2	June 28 – July 5, 2026	System Design and Database Design	Created system architecture, database structure, ERD, UI prototypes,	Completed

			and module designs for the Co-Curricular Management System.	
Sprint 3	July 6 – July 12, 2026	Role-Based Access Control (RBAC)	Developed user authentication, role permissions, and access restrictions for Admin, SSC President, Officer, and Student accounts.	Completed
Sprint 4	July 13 – July 19, 2026	Club Registration Portal and Student Club Membership Management	Developed club registration process, organization profiles, membership applications, and member management features.	Completed

Sprint 5	July 20 – July 26, 2026	Club Officer Elections and Event and Activity Logs	Implemented officer election management, candidate records, voting process, and recording of organization activities and events.	Completed
Sprint 6	July 27 – August 2, 2026	Attendance Tracker and Volunteer Hour Tracking	Developed attendance monitoring, QR code attendance feature, and student volunteer hour recording.	Completed
Sprint 7	August 3 – August 9, 2026	Club Achievement Records, Budget Request Management, Inter-Club	Created achievement documentation, budget request workflow, organization announcements, and	Completed

		Communication, and Club Directory	directory of registered clubs.	
Sprint 8	August 10 – August 15, 2026	Intelligent Reporting and AI-Based Activity Recommendations	Developed automated reports, activity insights, and AI-based recommendations based on student participation and interests.	Completed
Sprint 9	August 16 – August 21, 2026	System Testing and Quality Assurance	Conducted functional testing, security testing, usability testing, bug identification, and system improvements.	Completed
Sprint 10	August 22 – August	Final Integration and Documentation	Integrated all modules, performed final evaluation,	Completed

	27, 2026		prepared technical documentation, and finalized system presentation materials.	

3.1.4 Tool Stack

Table 5: Tool Stack

Development Component	Tool
Frontend	HTML 5, Tailwind, CSS, Laravel Blade
Backend	Laravel Framework
Database	MySQL
Programming Languages	PHP, JavaScript
API	REST APIs, API Integration, Postman
Authentication	Laravel Sanctum
Version Control	Git, Github
DevOps & Deployment	WSL, Ubuntu Linux

3.2 Architectural Design Decisions

The Co-Curricular Management System was carefully planned to ensure that its structure is organized, scalable, and easy to maintain. The architectural design was based on the needs of the users and the different functions of the system. Each module is designed to work together while still having its own specific purpose. This approach helps make the system easier to manage, update, and improve as the needs of the school and its users change.

The design also considers the proper flow of data between the different modules like Club Registration, Student Registration, Club Officer Election, Event and Activity Logs, Attendance Tracker, Club Achievement Records, Budget Requests, Inter-Club Communication, Volunteer Hour Tracking, and Club Directory. Each module is connected to the appropriate processes so that information can be properly recorded, updated, and accessed when needed. This helps ensure that the different parts of the system work together smoothly and that the data remains organized and consistent throughout the system.

3.2.1 Why Microservices? Justify the Choice over Monolithic Architecture

We chose Microservices Architecture because our proposed Co-Curricular Management System consists of several independent modules, including the Club Registration Portal, Student Club Memberships, Club Officer Elections, Event and Activity Logs, Attendance Tracker, Club Achievement Records, Budget Requests, Inter-club Communication, Volunteer Hour Tracking, and Club Directory. Each module performs a different function and can operate as a separate service. With microservices, each module can be developed, tested, updated, and maintained independently without affecting the rest of the system. This makes the development process more organized and reduces the possibility of system-wide failures whenever changes are made.

Compared to Monolithic Architecture, where all modules are combined into one large application, microservices provide better flexibility, scalability, and maintainability. If the school decides to add new features or improve an existing module in the future, developers only need to modify the specific service instead of rebuilding the entire system. This also allows multiple developers to work on different modules simultaneously, making development faster and more efficient. Because of these advantages, Microservices Architecture is the most suitable architecture for our Co-

Curricular Management System, especially since the system is expected to grow and continuously evolve as the needs of Bestlink College of the Philippines change.

3.2.2 Core Architectural Patterns Used

The Co-Curricular Management System uses architectural patterns to promote modularity, maintainability, scalability, and efficient data management.

This system follows a client-server architecture. Users access the application through a web browser, while the server handles the business logic, data processing, authentication, and communication with the database. This architecture helps organize data management and provides better control over system access.

The system also uses the Model-View-Controller (MVC) pattern, which is provided by the Laravel Framework. The Model manages the data, the View presents information to the users, and the Controller processes user requests. Separating these responsibilities helps keep the code organized and makes the system easier to maintain.

The system also uses API Architecture. This facilitates communication between system functionalities and external services and allows standardized data exchange. The Co-Curricular Management

System can integrate with AI-Based Activity Recommendations through an API, allowing the system to send the needed data and receive activity recommendations.

By combining these architectural patterns, the system achieves a more organized and maintainable structure. It supports regular operations, intelligent decision-making, and continuous development. This approach also helps improve code organization, system performance, and the overall management of co-curricular activities in the school.

3.3 Development, Operations, and Quality Assurance (QA) Methodology

The development, operations, and quality assurance (QA) methodology for the Co-Curricular Management System combines DevOps practices and software testing to make sure that the system is reliable, secure, and works properly. By connecting the development and operations processes, the team can develop, test, and deploy new features such as the AI-Based Activity Recommendations and Intelligent Reporting modules while maintaining the stability of the system.

This phase of the project focuses on making the development and deployment process more organized and reducing repetitive tasks. It also includes securing the database configuration during local development and

testing the system's web pages and API endpoints. The QA process uses different tools to check the backend functions and test the system's user interface. These tests help ensure that both the SSC administrative features and student-facing pages work properly when handling different amounts of data and user activities.

3.3.1 DevOps Toolchain

The Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations used DevOps to manage software development and deployment. This made the development and deployment. This made the development process more efficient and was able to manage its code and deployment workflow in a better way.

The Co-Curricular Management System used Git to keep track of changes made to the software applications made to the software applications code. Git helped the Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations to manage different versions of the application. It also helped to ensure the code was correct and could be changed back to a version if needed.

The system used GitHub to host its source code. It is a website where developers can work together on software projects. It helps developers manage their source code. The system found it helpful to use

GitHub for software development. The web application for the Co-Curricular Management System was deployed using Vercel. Vercel made it easy to publish the application after it was developed and tested. It can be also used to show the application to others and see how it worked.

The use of Git, GitHub, and Vercel creates an effective DevOps cycle which facilitated continuous development and collaboration, version, control, and deployment of the Co-Curricular Management System.

Table 6: DevOps Toolchain

Tools	Purpose
Git	Tracks changes in the source code and manages different versions of the application throughout the development.
GitHub	Serves as the repository for source code management, collaboration, and version control.
Vercel	Deploys and hosts web application for testing, demonstration, and implementation.

3.3.2 CI/CD Pipeline Design

The Co-Curricular Management System follows a Continuous Integration and Continuous Deployment (CI/CD) process. The CI/CD process helps manage code changes and deploy new versions of the application in an organized way.

First, during the development stage, new features and bug fixes are created and tested in the development environment. Once the changes have been checked and confirmed to be working properly, the updated code is committed to Git and pushed to the project's GitHub repository. The GitHub repository stores the different versions of the source code, making it easier for the team to track development progress and manage code changes.

After the updated code has been committed to the repository, the latest version of the web application is deployed using Vercel. This allows the updated application to become available for further testing and evaluation.

The CI/CD process used in the project is simple but effective. By using Git, GitHub, and Vercel together, the team can deploy updates more efficiently, track changes, support collaboration, and continuously improve the web application throughout the development process.

3.3.3 Testing Strategy

Testing will be conducted on the proposed Co-Curricular Management System to make sure that all functions work properly and meets the needs of the Supreme Student Council in terms of its functional and technical requirements. This will be done during the process of the development of the system, and all problems that arise will be solved before the final deployment of the system.

Below are the testing strategies that will be conducted:

Unit Testing: To verify whether the individual units and modules like Club Registration, Student Membership, Attendance Tracker, Inter-Club Communication, Club Directory, Intelligent Reporting, and AI-Based Activity Recommendations performed as required.

System Testing: To test the system's functionality, performance, reliability, and security.

Integration Testing: To test whether all units and modules communicate properly when integrated.

API Testing: To check the communication between the system and its connected services.

Security Testing: To ensure that the Co-Curricular Management System protects user accounts, student information, organization records, and other important data from unauthorized access and common security threats.

User Acceptance Testing (UAT): To allow the SSC to use the system and provide feedback based on their experience.

3.4 Innovation Framework

The development and implementation of the Co-Curricular Management System with Intelligent Reporting and AI-Based Activity Recommendations is guided by an innovation framework that focuses on using digital technologies to improve the management of student organizations and co-curricular activities. The framework aims to provide a more organized and efficient system that will help students, club officers, and the Supreme Student Council (SSC) manage organization-related activities, records, and participation.

While existing co-curricular management processes may rely on manual records, separate files, and traditional methods of monitoring student activities, the proposed innovation introduces a centralized web-based system that integrates important co-curricular processes. These include the Club Registration Portal, Student Club Memberships, Club

Officer Elections, Event and Activity Management, Attendance Tracking, Club Achievement Records, Volunteer Hour Tracking, and Club Directory. Through these integrated modules, information can be stored and accessed more easily by authorized users, reducing duplicate records and improving the overall organization of co-curricular information.

The innovation framework integrates Intelligent Reporting and AI-Based Activity Recommendations into the Co-Curricular Management System. Intelligent Reporting allows the system to generate organized reports based on information such as club memberships, student attendance, events, achievements, and volunteer hours. On the other hand, AI-Based Activity Recommendations can provide suggested activities to students based on their interests and previous participation. These features can help the SSC and student organizations understand participation patterns and provide activities that are more relevant to students.

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APPENDICES

APPENDIX A : DETAILED TECHNICAL DOCUMENTATION

1. System Architecture
2. Information Systems Integration
3. Application Design and Development
4. Database Schema and Data Management:
5. Network Configuration
6. Deployment and Infrastructure
7. Security Measures:
8. Testing and Quality Assurance
9. System Monitoring and Maintenance:
10. APIs and Integration Points:
11. User Documentation:
12. Known Issues and Troubleshooting
13. Version Control and Source Code Repository
14. DevOps and Continuous Integration/Continuous Deployment (CI/CD):
15. Licensing and Open Source Libraries:
16. Performance Metrics and Monitoring:

APPENDIX B : DOCUMENTS/URLS

1.Adviser Acceptance (signed)

2. Company Confirmation (signed)

3. Panel Evaluation and Signature (plus photo ops during defense)

4. Pilot Companies' Background with proofs of interviews

5. Link to codes (good for 3 years)

6. Link to documentation (good for 3 years)

7. Link to demo video guide (good for 3 years

APPENDIX C: CERTIFICATE OF ORIGINALITY

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This certifies that the research titled, **LOCAL GOVERNMENT UNIT 1: SOLID WASTE MANAGEMENT SYSTEM (COMPLAINT SUBMISSION, WASTE COLLECTION MANAGEMENT, REPORTING AND ANALYTICS)** has been checked and passed with a Turnitin similarity index of **10%.** Given this **3rd of June, 2025.**

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Librarian

JOHN PROS B. VALENCIA, RL, MLIS
Library Director

TC2421100

APPENDIX D: CERTIFICATE OF GRAMMARIAN

Name of Group Members/Individual:

1		4	
2		5	
3		6	

Title of the Project: EIS:

	Adviser	Grammarian
Name		
Designation		
Work Place Address		
Academic/ Professional Qualifications/ Membership		
Email Address		
Telephone		
Signature	*I hereby confirm that I have undertaken to supervise the project mentioned above and I do certify that I am not a member of the Project Examination Board (PEC)*	
	_____ *Date:*	_____ *Date:*

APPENDIX E : GRAMMARIAN CURRICULUM VITAE



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- Critical and Problem Solving
- Leadership
- Research Advising and Capstone Supervision

I hereby certify that the above information is true and correct to the best of my knowledge and belief

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APPENDIX F : TECHNICAL ADVISER CURRICULUM VITAE



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APPENDIX G: RESEARCH ADVISER CURRICULUM VITAE



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APPENDIX H : RESEARCHER'S CURRICULUM VITAE



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